

AVTAR SINGH

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SKILLS

- Languages: Python, Java, SQL, C
- AI/ML: LangChain, OpenAI API, NLP, RAG, Embeddings, Prompt Engineering, Sentence Transformers, Scikit-learn, Pandas, NumPy
- Frameworks/Tools: FastAPI, Flask, Streamlit, Docker, Git, Power BI, Tableau, MS Excel
- Databases: SQL, FAISS (Vector DB)
- Soft Skills: Critical Thinking, Strategic Planning, Time Management

PROJECTS

Multi-Agent Research Assistant | [GitHub](#) | LangChain, OpenAI, FastAPI, Agentic AI December 2025

- Built a production-style multi-agent AI research system using LangChain , enabling autonomous web search, content extraction, summarization, and report generation workflows.
- Designed specialized AI agents for search, retrieval, reasoning, report writing, and critique, orchestrated through shared memory and agent collaboration pipelines.
- Integrated LLM-powered retrieval and reasoning pipelines to generate structured research reports from live internet sources with contextual summarization and evaluation.

Multi-PDF Conversational AI Assistant | [GitHub](#) | LangChain, OpenAI, Hugging Face, Streamlit April 2025

- Developed a Retrieval-Augmented Generation (RAG) based conversational AI system enabling semantic question-answering across multiple PDF documents using LangChain and vector embeddings.
- Implemented document chunking, embedding generation, similarity search, and conversational memory pipelines using FAISS and Hugging Face embedding models for context-aware retrieval.
- Integrated LLM-powered response generation with Streamlit-based interactive UI to deliver accurate, context-grounded answers from uploaded documents in real time.

Deep Learning Time-Series Forecasting System | [GitHub](#) | Python, TensorFlow, LSTM, Data Analytics July 2024

- Built an end-to-end deep learning forecasting system using LSTM networks to analyze historical time-series data and predict future financial trends through sequential learning and temporal dependency modeling.
- Performed data preprocessing, normalization, and sliding-window sequence generation to optimize model training, improve forecasting consistency, and enhance prediction reliability across financial datasets.
- Designed model evaluation and visualization pipelines to compare predicted and actual trends using loss analysis, performance metrics, and graphical forecasting insights

Food Delivery Business Analytics Dashboard | [GitHub](#) | Power BI, SQL, Data Analytics June 2024

- Built an end-to-end business analytics dashboard in Power BI for a food delivery platform, analyzing revenue trends, customer behavior, operational metrics, and business performance insights.
- Performed data cleaning, transformation, modeling, and interactive KPI dashboard development using Power Query and Power BI to support data-driven decision-making and reporting workflows..

TRAINING

Database Developer (Academic Project / Internship) July 2025

- Architected and optimized a relational student-course management system using normalized database design, SQL procedures, and automated enrollment workflows to streamline academic operations.
- Developed complex SQL queries, joins, and transactional logic to enable efficient data retrieval, waitlist management, and scalable backend database processing.
- Implemented automated seat allocation and student promotion mechanisms, improving enrollment handling efficiency and ensuring data consistency across academic records.

CERTIFICATES

- Oracle Certified Data Science Professional, Oracle University | [Certificate](#) September 2025
- Oracle Certified Generative AI Professional, Oracle University | [Certificate](#) August 2024
- Oracle Certified AI Foundations Associate, Oracle University | [Certificate](#) February 2024

CO CURRICULAR ACTIVITIES

- Participated in a 24-hour inter-university CTF competition, Republic of Cyber Sentinels, working in a 3-member team and achieving a Top 5 position through effective collaboration and problem solving. January 2024

EDUCATION

Lovely Professional University Phagwara, Punjab
Bachelor of Technology in Computer Science Engineering Since August 2023

MGN Public School Jalandhar, Punjab
Intermediate September 2020 – April 2022